

# A Revisit on Covariant Approach in Relativistic Thermodynamics

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## ABSTRACT

This article seeks to address the challenge of defining thermal equilibrium covariantly. As a consequence, the question arises of whether thermodynamics can be approached in a covariant framework. The article analyses the topic of Relativistic Thermodynamics, focusing on the different points of view, especially in the problem of  $\tilde{T} = \gamma^a T$  and covariant forms of the first law of thermodynamics. With special reference to the question of how to define temperature in dynamical spacetime by studying the response and thermal behaviour of an Unruh-DeWitt detector as it travels through cosmological spacetimes. Also, through Nakamura's solution, this article manages to recover all other cases that had appeared in the covariant formulation of relativistic temperature, including Planck-Einstein, Ott-Kibble-Molle, Landsberg, and Doppler.<sup>a</sup>

## I. INTRODUCTION

Let us begin quoting[13] so we have a glimpse of what to expect:

"For a classical subject with no known practical application and little influence on current research fields, relativistic thermodynamics has aroused a remarkable amount of interest among physicists. Even more remarkable is the diversity of viewpoints presented. During the years 1965-1968, scores of papers on the subject were published by researchers in over half a dozen countries, giving almost ten different formulations. Yet the basic issue at stake was quite simple. It was merely a matter of choosing different definitions of the physical quantities involved. The trouble was, however, that each author's choice was so reasonable that it became rather difficult for him to see how others could possibly disagree. And thence, the controversy."

Despite this, there are still some things in which all authors agree, namely the invariance of entropy (associated with the number of microstates, which does not depend on the reference frame) and the pressure (because of the transformations of force and area). In total, we can classify the authors (or theories) into five different categories:

1. Classical authors (Planck), also supported by more recent ones (Penney, Staruszkiewicz, Pathria,

Eberly, Kujalski), defend that internal energy  $U$  and temperature  $T$ , measured from a reference frame moving at speed  $v$ , look like

$$U = \gamma (U_0 + v^2 P V_0) \quad (I.1)$$

$$T = \gamma^{-1} T_0 \quad (I.2)$$

where the subscript 0 refers to the rest-frame values,  $P = P_0$  is the pressure,  $V$  is the volume and the relativistic factor is

$$\gamma = (1 - v^2)^{-1/2}$$

This problem serves as an excellent illustration of covariance in defining nonlocal quantities using integration. The classical expression for energy density is

$$u(\omega)d\omega = A\omega^2 d\omega / [\exp(\hbar\omega/kT) - 1].$$

Eberly and Kujawski, by integrating the expression in a moving reference frame, obtained the total energy of a photon cavity as

$$U = \gamma U_0 \left( 1 + \frac{1}{3} v^2 \right).$$

This agrees with the value given by Eq. (I.1)

$$U = \gamma (U_0 + v^2 P V_0)$$

because

$$P_0 = \frac{1}{3} U_0 / V_0.$$

Once again, the integration is performed at a constant time in the observer's frame. This destroys covariance and energy-momentum; thus, the obtained does not form a 4-vector.

To obtain a covariant definition of total energy, It is first observed that the number of states with wave vector

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**Note:** All constants possible ( $c, \hbar, k_B \dots$ ) equal to 1.

**Disclaimer:** In this article, we choose to work in the Pozas gauge.

between  $k$  and  $k + dk$  is proportional to  $Vd^3k$ , and the probability that such a state is occupied is given by Bose statistics, so that.

$$n(k)d^3k = A'Vd^3k/[\exp(\hbar\omega/kT) - 1]$$

with

$$A' = 2/(2\pi)^3 = c^3A/4\pi\hbar.$$

Multiplying Above Eq. by  $\hbar\omega$  and integrating over angles gives the energy of all photons with frequency  $\omega$ . Thus

$$U(\omega)d\omega = AV\omega^3d\omega/[\exp(\hbar\omega/kT) - 1],$$

Now consider the problem relativistically. We note that  $(k, i\omega)$  forms a 4-vector, denoted by  $\omega_i$ . From Sec. III, we know that  $\exp(\hbar\omega/kT)$  is to be replaced by  $\exp(-\hbar\beta_i\omega_i)$ , so that the denominator of above Eq. is a scalar. The numerator must also be a scalar since  $n(k)d^3k$  is the number of photons within a range of wave vectors. Thus we replace  $Vd^3k$  by  $V_id^3\omega_i$ . Multiplying the result by  $\hbar\omega_i$  and integrating over  $\omega_i$  we have the total energy-momentum:

$$E_j = A' \int \omega_j V_i d^3\omega_i / [\exp(-\hbar\beta_i\omega_i) - 1].$$

This obviously reduces to the classical expression in the rest frame, where  $V_{1,2,3}$  and  $\beta_{1,2,3}$  are zero.

$$E_j = A' \int \omega_j V_0 \gamma (d^3k - v d\omega dk_x dk_y) / \{\exp[\gamma\hbar(\omega - vk_z)/kT_0] - 1\} \quad (1)$$

We note that integration of the above Eq. over all angles while keeping  $k$  constant entails the constraint.

$$k_x^2 + k_y^2 + k_z^2 = \omega^2,$$

Which is in fact

$$\omega_i\omega_i = 0.$$

The above Equation is a scalar relation, hence Eq. E<sub>j</sub> is valid in all frames. We can integrate over angles.

As we did before. Changing to angle variables, we have where

$$E_j = A' \int \omega_j V_0 (dx - vxdx) k^2 dk d\phi / \quad (2)$$

$$\{\exp[\gamma\hbar\omega(1 - vx)/kT_0] - 1\}, \quad (3)$$

$$x = \cos\theta.$$

Putting  $s = \gamma\omega(1 - vx)$ , integrating over  $\phi$ , and using

$$2\pi A' \hbar/c^3 = \frac{1}{2}A,$$

We get the total energy.

$$U = (1/2\gamma^3) \int (1-vx)^{-3} dx \int AV_0 s^3 ds / [\exp(\hbar s/kT_0) - 1].$$

The last integral gives  $U_0$ , and we see

$$U = U_0 \int \frac{1}{2} \gamma^{-3} (1 - vx)^{-3} dx \\ = U_0 (1 - v^2)^{-1/2}$$

This is what a covariance would demand. **whether a system is free or not, its energy-momentum is always a four-vector provided one performs covariant summation at constant time in the frame of the system .**

2. Ott, Kibble, and Møller accept the transformation for energy Eq.(I.1) but argue that the appropriate transformation for temperature can be written as

$$T = \gamma T_0 \quad (I.3)$$

3. Arzeliès, Gamba, and Bors, who prefer Eq. I.3 as the transformation of temperature and

$$U = \gamma U_0 \quad (I.4)$$

As the transformation of energy.

4. Landsberg and van Kampen argue that temperature is a scalar quantity, so it is not different when measured from a reference frame to the rest of the one.
5. Rohrlich, Balescu, or Sewell have different views, among which exists that of temperature being a quantity that only makes sense to define in the rest frame.

As one can see, there are a lot of points of view to discuss about. Let us begin seeing the different arguments and discussions.

## II. THE OLD EINSTEIN-PLANCK THEORY

Let us see how the two Equations I.1 and I.2 are derived from the simple model of a fluid in a container of rest-volume  $V_0$ . In the rest frame of the fluid, its stress-energy tensor is

$$T_0^{\mu\nu} = \begin{pmatrix} \rho_U & 0 \\ 0 & P \end{pmatrix}$$

where  $\rho_U$  is the energy density and  $P$  is the pressure. One can note that we have used a 1 + 1 - D model, but the 3 + 1-D case is straightforward.

Given that the volume in the rest frame is  $V_0$ , the energy and momentum of the container are

$$U^0 = \int_{V_0} T_0^{00} dx_0 = \int_{V_0} \rho_U dx_0 = \rho_U V_0$$

$$p^0 = \int_{V_0} T_0^{01} dx_0 = 0$$

Now, if we observe the same container from a reference frame moving at speed  $v$ , the volume shrinks by a factor of  $\gamma^{-1}$  (the usual length contraction), and the stress-energy momentum tensor is

$$T^{\mu\nu} = \Lambda^\mu_\rho T_0^{\rho\sigma} \Lambda^\nu_\sigma = \gamma^2 \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix} \begin{pmatrix} \rho_U & 0 \\ 0 & P \end{pmatrix} \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}$$

$$= \gamma^2 \begin{pmatrix} \rho_U + v^2 P & -v(\rho_U + P) \\ -v(\rho_U + P) & P + v^2 \rho_U \end{pmatrix}$$

Therefore, the energy and momentum that a moving observer sees are

$$U = \int_V T^{00} dx = \int_{V_0 \gamma^{-1}} \gamma^2 (\rho_U + v^2 P) dx = \gamma (U_0 + v^2 P V_0) \quad (I.5)$$

$$p = \int_V T^{01} dx = - \int_{V_0 \gamma^{-1}} v \gamma^2 (\rho_U + P) dx = -v \gamma (U_0 + P V_0) \quad (I.6)$$

where  $U_0 = \rho_U V_0$ .

One of the first objections one finds is that the quantities  $U, p$  do not form a four-vector in the sense that  $(U, p)^\mu \neq \Lambda^\mu_\nu (U^0, p^0)^\nu$ . Essentially, the problem stems from the fact that when calculating the energy as  $U = \int_V T^{00} dx$ , we are doing it at a constant  $t$  that is constant in the reference frame of the moving observer, instead of computing it at the transformation of the time in the rest frame,  $t_0$ . Therefore, we are measuring two different quantities, which do not need to be related by a Lorentz transformation. Appropriate consideration of this subtlety will lead to Nakamura's Formalism, which will be discussed later.

The temperature transformation is obtained using similar arguments, and in particular, making the same "mistake" as in the transformation of the energy. The transformation of work is

$$dW = P dV = P_0 d(\gamma^{-1} V_0) = \gamma^{-1} dW_0 - P V_0 \gamma v dv$$

since  $d\gamma = \gamma^3 v dv$ .

Considering Internal energy as  $U$  and change in momentum as  $dp$  (we are considering the speed of light  $c=1\text{m/s}$ , so  $U$  has the dimension of Energy and  $U/c$  has

the dimension of momentum but magnitude as  $U$ ), The heat flow is calculated as

$$dQ = dU + dW + v \cdot dp \quad (I.8)$$

Where the last term has been inserted to take into account the energy that is needed to accelerate the system (or the observer) to speed  $v$ , other works [14] argue that this term is needed to compensate for the gain or loss of radiation energy, which would result in acceleration. Anyway, plugging in the definitions of  $U$  and  $W$ , we obtain

$$dQ = \gamma^{-1} (dU_0 + dW_0) = \gamma^{-1} dQ_0$$

And finally, given that the entropy is a scalar and if we assume  $T dS = dQ$ , then we have Eq. I.2

$$T = \gamma^{-1} T_0$$

As we can see, there are a few "sketchy" things in this derivation. First, we have already seen that energy and momentum do not form a four-vector because we measure different things in different frames rather than asking how a specific thing looks when seen from different frames. Also, a correction has to be inserted in the first law of thermodynamics Eq. I.8 Even energy conservation is not apparent, although it can be inserted, and then the Formalism is consistent, but still, it looks like you need to insert too many things, and one may believe that a more "natural" approach must exist.

This point of view, however, obtained a lot of support since another totally independent way of arguing brought the same results. This argument relied on the assumption that the equilibrium distribution function of a system is invariant under Lorentz transformations, i.e., it is a scalar. Therefore, if all particles have energy  $u$  and momentum  $p$ , this amounts to say that

$$f(u, p) = f'(u', p'),$$

where  $e^\mu = (u, p)^\mu = \Lambda^\mu_\nu (u_0, p_0)^\nu$ . Furthermore, the equilibrium distribution depends only on energy and not on momentum via the factor  $\beta u$  (where  $\beta = T^{-1}$ ). Therefore, in order to "covariantize" this factor, one should transform it into a scalar product  $\beta \cdot e = \beta_\mu e^\mu$ , where  $\beta_0^\mu = (T^{-1}, 0)$  in the rest frame. Transforming this vector and comparing the zeroth components, one arrives at the conclusion that.

$$\beta^0 = \gamma \beta_0^0 \Rightarrow T = \gamma^{-1} T_0$$

Also, if one computes the energy and the momentum

$$E^\mu = \int e^\mu f(\beta_\nu e^\nu) dp dV$$

One obtains Eqs. I.5 and I.6, but again, we are making the same error, namely measuring at constant  $t$  instead of asking what the distribution at constant  $t_0$  looks like in the new frame.

### III. COVARIANT THEORIES

#### A. First Covariant Theories:

Rohrlich made the first attempts to make a covariant formulation of thermodynamics, thinking of classical results as being the zeroth components of covariant quantities. Beginning with the  $T dS$  equation

$$T_0 dS = dU_0 + P dV_0$$

Multiplying both sides by the velocity four-vector  $v^\mu = \gamma(1, v)$  we get the covariant equation

$$T^\mu dS = dU^\mu + P dV^\mu$$

where  $T^\mu = v^\mu T_0$  and  $V^\mu = v^\mu V_0$ . In this way, energy and momentum can be defined in such a way that they form a four-vector

$$p^\mu = \int_V T^\mu_\nu dV^\nu$$

and the temperature (given the transformation of  $v^\mu$ ), would transform as

$$T = \gamma T_0$$

However, this formulation does not allow to study heat exchange when the two bodies are moving with different velocities, given that  $dQ^\mu = v^\mu (T_0 dS)$ , and we cannot equate this expression to  $dQ^\mu_A - dQ^\mu_B$  unless both systems have the same velocity.

#### B. More serious covariant theories

Considering the following is not a proposed relativistic generalization of the First Law, it is the First Law itself; we can proceed with the concept that the 4-momentum  $p_\mu = (-U, \vec{P})$  transforms as a 1-form, we can read off from the expression.

$$dS = \beta dU - \beta \vec{v} \cdot \vec{P}$$

(where  $\beta = 1/T$ ) that the object  $(\beta, \beta \vec{v})$  transforms as a vector, which we might call the *inverse 4-temperature*. We can then write the First Law in the manifestly covariant form

$$dS = -\beta^\mu dp_\mu$$

where  $\beta^\mu$  is the inverse 4-temperature and  $p_\mu$  is the 4-momentum. The physical meaning of inverse 4-temperature is (minus) the rate of change of entropy with 4-momentum.

We can rewrite the inverse 4-temperature as

$$\beta^\mu = (\beta, \beta \vec{v}) = \left( \frac{\beta}{\gamma(\vec{v})}, \beta \gamma(\vec{v}), \beta \vec{v} \gamma(\vec{v}) \right) \equiv \beta_R v^\mu$$

where  $v^\mu = (\gamma(\vec{v}), \vec{v} \gamma(\vec{v}))$  is the 4-velocity and  $\beta_R = \beta/\gamma(\vec{v})$  is the *inverse rest temperature*. The inverse rest temperature is a scalar and can be defined directly in terms of the inverse 4-temperature via

$$\beta_R^2 = -\beta^\mu \beta_\mu.$$

To retrieve the physical interpretation, we can see:

$$\beta_R dM = \frac{\beta_R}{2M} d(-p^\mu p_\mu) = -\frac{\beta_R p^\mu}{M} dp_\mu = -\beta^\mu dp_\mu = dS$$

Eberly and van Kampen found their way out of here. Multiplying  $dQ^\mu = v^\mu (T_0 dS)$  by  $\beta_\mu = \beta v_\mu$ , one obtains

$$dS = -\beta_\mu dQ^\mu$$

If we take this equation as fundamental (as Eberly and van Kampen did), we have no relation between heat and velocity; two systems  $A, B$  having identical rest temperature  $T$  are in relative motion, with  $B$  having velocity  $+v$  relative to  $A$  along the  $x$  axis. Let's analyze two systems that come into thermal contact for some small period of time, so  $dQ^\mu_A = -dQ^\mu_B$ . The entropy change of this process is

$$dS = dS_A + dS_B = dQ_\mu (\beta^\mu_B - \beta^\mu_A)$$

In  $A$ 's rest frame,  $B$  has temperature  $T/\gamma(v)$ , and so it is entropically favorable for heat to flow from  $A$  to  $B$ . But in  $B$ 's rest frame,  $A$  likewise has temperature  $T/\gamma(v)$ , and it is thermodynamically favorable for heat to flow from  $B$  to  $A$ . At a glance, this is a contradiction.

**The resolution of the paradox comes from noting that the statement 'energy is transferred but momentum is not' is not Lorentz-invariant.** In  $A$ 's rest frame, consider a transfer of some small amount of energy  $\Delta U$  from  $A$  to  $B$  without any additional transfer of momentum. The 4-momentum transferred  $\Delta p_\mu$  has components as follows:

$$\Delta p_\mu = (-\Delta U, 0, 0, 0).$$

This transfer increases total entropy by

$$\Delta S = \Delta U \left( \frac{\gamma(v)}{T} - \frac{1}{T} \right) = \frac{\Delta U}{T} (\gamma(v) - 1) > 0$$

since the temperature of  $B$  is  $T/\gamma(v)$ . Now suppose we instead transfer a quantity of 4-momentum  $p'$  that, in  $B$ 's rest frame, is purely a transfer of energy  $\Delta U$ . In  $A$ 's rest frame, this transfer also involves a transfer of momentum: we have

$$p'_\mu = (-\gamma(v)\Delta U, v\gamma(v)\Delta U).$$

The inverse 4-temperatures of  $A$  and  $B$  in  $A$ 's rest frame are, respectively,

$$\beta_A = (1/T, 0, 0, 0)$$

and

$$\beta_B = (\gamma(v)/T, v\gamma(v)/T, 0, 0)$$

and so the change of entropy (continuing to calculate in  $A$ 's rest frame) is

$$\begin{aligned} \Delta S &= \gamma(v)\Delta U \left( \frac{1}{T} - \frac{\gamma(v)}{T} \right) - v\gamma(v)\Delta U \left( 0 - \frac{v\gamma(v)}{T} \right) \\ &= \frac{\gamma(v)\Delta U}{T} (1 - (1 - v^2)\gamma(v)) \\ &= \frac{\Delta U}{T} (\gamma(v) - 1). \end{aligned}$$

If the two bodies are in thermal equilibrium,  $dS = 0$  and therefore  $\beta_A^\mu = \beta_B^\mu$ , which implies equal velocities and rest-frame temperatures. We can conclude that **unless the bodies have equal velocities and rest frame temperatures, heat exchange must be irreversible.**

There was still a point of disagreement between Eberly and van Kampen. They both accepted  $dS = -\beta_\mu dQ^\mu$  as correct but still considered that the temperature transformed differently. While Eberly argued, from the form of  $\beta^\mu = \beta_0 v^\mu$  that Eq. I.2 was the correct transformation, van Kampen interpreted  $dS = \beta v^\mu dQ_\mu$ , and therefore the temperature was a scalar which does not transform under Lorentz transformations.

There was also another argument for the definition of  $T$  as a scalar. Namely, for two systems in relative motion with the same rest frame temperature in thermal contact, an observer at rest in each system would see heat flowing inside or outside its system, therefore creating a paradox. However, this argument is only valid if  $T$  is the only property that tells the direction of the heat flow, and the paradox does not appear with a suitable definition of the second law.

### C. Why define a temperature at all?

Landsberg changed his point of view and joined others like Sewell, arguing that a temperature cannot be defined when observing from a moving frame. They investigated the response of a particle detector coupled to a field in the conformal vacuum of an FLRW universe. Within the

Unruh-DeWitt particle detector model in Quantum Field Theory in curved spacetime (QFTCS), consider an idealized two-level quantum mechanical measuring device, or detector, as it travels along a given trajectory through the expanding universes of de Sitter space and an FLRW spacetime with a scale factor of the form  $a(t) \sim \cosh(\lambda t)$

In stationary spacetimes, for instance, static black holes, in the event horizon (the two-dimensional null hypersurface formed where null and timelike Killing vectors coincide), the four-acceleration on the horizon still diverges.

This ensures that the surface gravity remains regular on the horizon and leads to a consistent notion of temperature  $T = \kappa/2\pi$ <sup>1</sup> on the event horizon of a black hole, known as *Hawking temperature*. **Although the classical notion of surface-gravity in *General Relativity*, defined as it is through the geometric properties of the spacetime, Killing vectors persist only in stationary spacetimes**, which questions the dynamical spacetimes such as a black hole with an evolving horizon or an expanding universe. **Without a well-defined definition of a Killing horizon, the framework for physical quantities such as four-acceleration, surface gravity as well as temperature doesn't fit well**.. Then, we should require some alternative prescription for surface gravity in dynamic spacetimes. To better understand the thermal behavior of the detector as it interacts with the quantum field and use this to compare competing definitions of surface gravity and temperature, one can refer to the approaches of Hayward-Kodama, Ashtekar, et al., Fodor et al., and Nielsen-Visser.[2]

## IV. PARTICLE DETECTOR THEORY

As with Refs. [3, 4], in the particle detector model we are considering a switching function  $\chi$  which controls how the interaction with the quantum field is turned on and off.. While the switching function is initially assumed to be smooth, we will see later that it approaches a sharp-switching step function when an appropriate limit is taken, provided that the quantum state of the field is Hadamard.

Suppose then that the particle detector travels along a world line  $x^\mu(\tau)$ , where  $\tau$  is the proper time of the detector. We model the interaction between the detector and the real-valued quantum field  $\hat{\phi}(x)$  via the interaction Hamiltonian

$$H_{int} = c\chi(\tau)\hat{m}(\tau)\hat{\phi}(x), \quad (4)$$

<sup>1</sup> where now the surface gravity is  $\kappa \sim V \times A$  where the red-shift factor is  $V$ , and  $A$  is the magnitude of the four-acceleration which serves to shift the locally applied force (acceleration) required to keep a point particle in place at some radius  $r$  to infinity through, what is often referred to as, an 'infinitely-long massless string' [1].

where  $c$  is a coupling constant assumed to be small, so that we can treat the system as a perturbation around the free Hamiltonian;  $\hat{m}(\tau)$  is the detector's monopole moment operator; and we choose  $\hat{\varphi}(x)$  to be massless.

Before the detector and the quantum field interact, assume that the field  $\hat{\varphi}(x)$  is in some initial Hadamard state  $|\Phi\rangle$  on a given background, while the detector is in its ground state  $|E_0\rangle$ . During the interaction, the field  $\hat{\varphi}(x)$  transitions from its initial state  $|\Phi\rangle$  to a final state,  $|\Phi'\rangle$ , while the detector undergoes a transition from ground state  $|E_0\rangle$  to excited state  $|E\rangle$ , also depends on the trajectory of the detector in the given spacetime.

When the detector leaves its ground state, the eigenvalue for the state  $|E\rangle$  may be positive or negative. In the case where  $E > E_0$ , the interpretation is that the detector has absorbed field quanta, while for  $E < E_0$ , the detector has emitted field quanta. To first order in perturbation theory (assuming that  $c$  is small), the probability for this transition is For sufficiently small  $c$ , the amplitude for such a transition,

$$A = -ic\langle E, \Phi' | \int_{-\infty}^{\infty} d\tau \chi(\tau) \hat{\mu}(\tau) \hat{\varphi}(x) | \Phi, E_0 \rangle. \quad (5)$$

Using the formula for the evolution of the moment operator  $\mu(\tau) = e^{iH_f\tau} \mu(0) e^{-iH_f\tau}$  where  $H_f$  is the free-field Hamiltonian, such that  $H_f|E\rangle = E|E\rangle$  and  $H_f|0\rangle = 0$ , gives  $H_0|E_0\rangle = E_0|E_0\rangle$ , i.e.

$$A = -ic\langle E | \mu(0) | E_0 \rangle \int_{-\infty}^{\infty} d\tau e^{i(E-E_0)\tau} \chi(\tau) \langle \Phi' | \hat{\varphi}(x) | \Phi \rangle. \quad (6)$$

Now, let us think of the detector and field as being in their ground state  $|E_0\rangle \otimes |\Phi\rangle$  at some initial time  $\tau_0$ . The detector has been switched on at time  $\tau_0$  and off at some later time  $\tau$ , the probability of the detector being observed in an excited state  $|E\rangle$  is obtained by first squaring the modulus of the amplitude (6), before summing over  $E$  and all possible fields  $\Psi$  [5, 6] to find

$$P(\omega) = c^2 |\langle E | \hat{m}(0) | E_0 \rangle|^2 \mathcal{F}(\omega), \quad (7)$$

where the *response function*<sup>2</sup>

$$\begin{aligned} \mathcal{F}(\omega) \equiv & 2 \lim_{\epsilon \rightarrow 0^+} \Re \int_{-\infty}^{\infty} du \chi(u) \times \\ & \times \int_0^{\infty} ds \chi(u-s) e^{-i\omega s} W_\epsilon(u, u-s), \end{aligned} \quad (8)$$

where the energy gap  $\omega \equiv E - E_0 = \Delta E$ , and  $W_\epsilon(x, x')$  is the Wightman Green function for the massless scalar wave equation. Eq. (7) depends only on the structure of the detector, which has been modeled as a monopole. Hence, one can refer to  $\mathcal{F}(\omega)$  as the transition probability.

<sup>2</sup> Here, we have chosen  $u = \tau$  and  $s = \tau - \tau'$  in the region where  $\tau > \tau'$ , while  $u = \tau'$  and  $s = \tau' - \tau$  in the region  $\tau' > \tau$ .

For a field in the *conformal vacuum*<sup>3</sup>, the Wightman two-point function, is<sup>4</sup>

$$W_\epsilon(x; x') = \frac{1}{(2\pi)^2} \frac{1}{a(\eta)a(\eta')} \frac{1}{-|\eta - \eta' - i\epsilon|^2 + |\mathbf{x} - \mathbf{x}'|^2}, \quad (9)$$

the details can be found in Appendix A.

The Wightman two-point function is regular except when  $x = x'$ . following the Hadamard singularity structure and requires a regularisation procedure to ensure meaningful results, according to ‘**Feynman prescription**’ of inserting a small parameter  $i\epsilon$  to regularize  $W_\epsilon(x; x')$  as a well-defined distribution giving unphysical results when used to reproduce the thermal Unruh spectrum seen by an accelerated observer.[7]

It was shown in Refs. [4, 8] that the regularisation (9) can be used so long as the switching functions  $\chi(\tau)$  are assumed initially to be smooth and of compact support with a sharp-switching limit where the response function (8) diverges at this limit, the rate of response remains finite so that it is this *transition rate*, which we interpret as the rate of particle detection, that will be the quantity of interest. It was shown in Ref. [4] and in Appendix B that the sharp-switching limit of the transition rate for FLRW Spacetime agrees with **Schlicht’s regularisation** to leading order and is given by

$$\dot{\mathcal{F}}_\tau(\omega) = \frac{1}{2\pi^2} \int_0^{\Delta\tau} ds \left( \frac{\cos \omega s}{\sigma^2(\tau, s)} + \frac{1}{s^2} \right) + \frac{1}{2\pi^2 \Delta\tau} - \frac{\omega}{4\pi}, \quad (10)$$

where  $\Delta\tau \equiv \tau - \tau_0$  is the detection time and the trajectories<sup>5</sup>  $\Delta x \equiv x(\tau) - x(\tau - s)$  are encoded in the geodesic distance  $\sigma^2(\tau, s) \equiv a(\tau)a(\tau - s)(\Delta x)^2$ .

The temperature of the detector can be on the detailed balance form of the Kubo–Martin–Schwinger (KMS) conditions, Ref. [9] to implement the thermal behavior of an Unruh–DeWitt detector, as it passes along a given trajectory in an FLRW spacetime. To arrive at a suitable temperature estimator, we first define the excitation to de-excitation ratio

$$\bar{\mathcal{R}} \equiv \frac{\mathcal{F}(\omega)}{\mathcal{F}(-\omega)} \quad (11)$$

and if there exists a constant  $T$  that satisfies the detailed-balance form of the KMS condition, then

$$\bar{\mathcal{R}} = e^{-\omega/T}, \quad (12)$$

<sup>3</sup> By ‘conformal vacuum’ means a mode decomposition based on the conformally-flat nature of the FLRW metric which reduces to the flat space mode decomposition when  $a(t) = 1$ .

<sup>4</sup> The spatial trajectories are encoded in  $\mathbf{x} = \mathbf{x}(\tau)$  while  $\eta = \eta(\tau)$  is the conformal time parameter and  $\tau = u$  is proper time throughout. Both spatial and temporal components are contained within the parameter  $x = x(\tau)$  while the ‘ $\eta$ ’ notation signifies  $x' = x(\tau - s)$ ,  $\eta' = \eta(\tau - s)$ , etc. The scale factor is given by  $a$ .

<sup>5</sup> Explicitly,  $(\Delta x)^2 = -(\eta(\tau) - \eta(\tau - s))^2 + (\mathbf{x}(\tau) - \mathbf{x}(\tau - s))^2$ .

and we can identify  $T = \bar{T}_{\text{EDR}}$  with the temperature of the detector which is given by

$$\bar{T}_{\text{EDR}} = -\frac{\omega}{\ln \mathcal{R}}. \quad (13)$$

For the static detector in the limit of finite detection times,  $\bar{T}_{\text{EDR}}$  becomes dependent on the energy gap and equals the locally measured Hawking temperature, but this dependence can be sufficiently weak so that it remains a suitable temperature estimator for the detector, see Ref. [10].

A complication arises from the fact that the response function diverges in the sharp-switching limit. In this case, instead of the ratio of excitation to de-excitation probabilities as above, we take instead the ratio of the rates

$$T_{\text{EDR}} = -\frac{\omega}{\ln \mathcal{R}}, \quad \mathcal{R} = \frac{\dot{F}_{\tau}(\omega)}{\dot{F}_{\tau}(-\omega)}. \quad (14)$$

As shown in Refs. [3],[23] this definition gives the expected  $T_{\text{EDR}} = T_{\text{loc}}$  in the limit of infinite detection time for a static detector. When Eq. (14) is a constant or approximately constant function of energy gap  $\omega$ , we will say the detector has thermalized at a temperature of  $T_{\text{EDR}}$ .

## V. MOTION OF THE DETECTOR

Consider an Unruh-DeWitt detector explained in the previous section and C, moving in Minkowski spacetime in a two-dimensional spatial plane with constant square of magnitude of four acceleration  $a_{\mu}a^{\mu} = a^2$ , where  $a^{\mu} = d^2x^{\mu}/d\tau^2$ , and constant magnitude of a timelike proper-time derivative of four-acceleration  $(da_{\mu}/d\tau)(da^{\mu}/d\tau)$ , and having component of the four-velocity,  $dy/d\tau = w$ , a constant, a detector moving along the following world-line (one of the components of the 4-velocity,  $w = dy/d\tau$  is constant)

$$x^{\mu}(\tau) = \left( \frac{a}{\alpha^2} \sinh(\alpha\tau), \frac{a}{\alpha^2} \cosh(\alpha\tau), w\tau, 0 \right), \quad (15)$$

$$\alpha = \frac{a}{\sqrt{1+w^2}} > 0$$

This is the motion of a charge in a uniform electric field when, in the frame of the charge, there is both an electric and a magnetic field.

The magnitude of the Fermi-Walker derivative of the acceleration is

$$|Da| = \left( D_{\mu}^{(F)}[a] D^{\mu(F)}[a] \right)^{\frac{1}{2}} = \frac{a^2 w}{\sqrt{1+w^2}}$$

The parameter  $\eta = |Da|/a^2$  is less than one. In a Fermi-Walker frame moving with the detector, the direction

of the acceleration rotates at a constant rate  $\eta$  around a great circle. For a circular trajectory where  $\eta > 1$ , replace the uniform electric field with a uniform magnetic field and take charge moving so that in the frame of the charge, there is both a magnetic and an electric field.

For  $w = 0$ , the trajectory is

$$x^{\mu} = x^{\mu}(\tau) = (a^{-1} \sinh(a\tau), a^{-1} \cosh(a\tau), 0, 0) \quad (16)$$

Which is a trajectory of a detector moving along a spatially straight line along the  $x$  direction with a constant magnitude of four-acceleration  $a_{\mu}a^{\mu} = a^2$ .

Instead of the component of the four-velocity  $w = dy/d\tau = \text{const.}$ , if we consider the component of the three-velocity  $dy/dt = v$  to be constant, the response of the detector would have been completely equivalent to that of a detector which is moving along a spatially straight line with constant magnitude of four-acceleration, as such observers can be related to the ones moving along the trajectory by Lorentz transformations.

## A. MOTION OF THE DETECTOR IN THE MINKOWSKI VACUUM

The positive frequency Wightman Green function for a massless scalar field for (15) is: [C]

$$W^{+}(\Delta\tau) = -\frac{\alpha^4}{16\pi^2 a^2} \left[ \sinh^2 \left( \frac{\alpha\Delta\tau}{2} - \frac{i\epsilon\alpha^2}{a} \right) - \frac{w^2\alpha^4}{4a^2} \Delta\tau^2 \right]^{-1},$$

Here, we have absorbed a positive function of  $\tau$  and  $\tau'$ ,  $[\sinh(\alpha\tau) - \sinh(\alpha\tau')]/[\sinh(\Delta\tau/2\alpha) \cosh(\Delta\tau/2\alpha)]$ , into  $\epsilon$ . For  $w = 0$ , we have  $\alpha = a$  and immediately convert to the positive Wightman Green function of a detector moving along a spatially straight line in the  $x$ , with constant magnitude of the four acceleration :

$$W^{+(1d)}(\Delta\tau) = -\frac{a^2}{16\pi^2} \left[ \sinh^2 \left( \frac{a\Delta\tau}{2} - i\epsilon a \right) \right]^{-1}$$

Where The (1d) index is used two, distinguish the Wightman Green function or response function calculated for the spatially one-dimensional trajectory and the index (2d) for the quantities associated with the spatially two-dimensional trajectory. The transition probability per unit proper time for the detector following trajectory (16) is the usual black body excitation rate,

$$p_{\Delta\tau} = m^2 \sum_E |q_{10}|^2 \mathcal{F}^{(1d)}(E)$$

$$\text{where} \quad \mathcal{F}^{(1d)}(E) = \frac{\Delta E}{2\pi [e^{2\pi\Delta E/a} - 1]}$$

Where  $\mathcal{F}$  is used to denote the response function.  $\Delta E = E - E_0$ . The excitation rate of an accelerated detector coupled to the field  $\varphi$  in the state  $|0_M\rangle$  is the same as that of a detector, unaccelerated, at rest in a bath of thermal radiation at temperature  $T = 1/\beta = a/(2\pi)$ .

In the "infrared limit"  $\Delta E \ll 1$ , the black body excitation rate has the following dominant behavior:

$$\mathcal{F}^{(1d)}(E) \sim \frac{1}{2\pi\beta}$$

In the "ultraviolet limit"  $\Delta E \gg 1$ , the black body excitation rate has the following dominant behavior:

$$\mathcal{F}^{(1d)}(E) \sim \frac{\Delta E}{2\pi} e^{-\beta\Delta E}$$

The response function per unit proper time of the detector following the trajectory (15) for nonrelativistic velocities  $v_y \ll 1$

$$v_y = \frac{dy}{dt} = \frac{w}{\sqrt{1+w^2} \cosh(\alpha\tau)}$$

or  $w \ll 1$ . For ultra-relativistic velocities  $v_y \rightarrow 1$  or  $w \rightarrow \infty$ , the response function of the detector (16) is suppressed. Namely, the Unruh effect is suppressed as

$$\mathcal{F}^{(2d)}(E) \sim \frac{\Delta E}{2\pi [e^{2\pi\Delta E/a} - 1]} \frac{1}{w^4}$$

The Wightman Green function in the leading order for  $w \ll 1$  reduces

$$\begin{aligned} W^{+(2d)}(\Delta\tau) &= (1 - 2w^2) W^{+(1d)}(\Delta\tau) \\ &- \frac{a^2}{16\pi^2} \left[ \sinh(a\Delta\tau - 2i\epsilon a) \left( \frac{a\Delta\tau}{4} - i\epsilon a \right) + \frac{a^2\Delta\tau^2}{4} \right] \times \\ &\left[ \sinh^4 \left( \frac{a\Delta\tau}{2} - i\epsilon a \right) \right]^{-1} w^2 + \mathcal{O}(w^4). \end{aligned}$$

For the response function per unit proper time, we use the following identity[11]:

$$\sinh x = x \prod_{k=1}^{\infty} \left( 1 + \frac{x^2}{k^2\pi^2} \right)$$

Calculating the integral

$$\mathcal{F}(E) = \int_{-\infty}^{\infty} d(\Delta\tau) e^{i(E-E_0)\Delta\tau} W^+(\Delta\tau, r),$$

Where we found,

$$\mathcal{F}^{(2d)}(E) = \mathcal{F}^{(1d)}(E) + \mathcal{F}^a(E)w^2 + \mathcal{O}(w^4)$$

where

$$\begin{aligned} \mathcal{F}^a(E) &= -\frac{e^{\beta\Delta E} \Delta E^2}{12\pi\beta [e^{\beta\Delta E} - 1]^2} \times \\ &\left[ \frac{8\pi^2}{\Delta E^2} + 9\beta^2 - \beta\Delta E \left( \frac{4\pi^2}{\Delta E^2} + \beta^2 \right) \frac{e^{\beta\Delta E} + 1}{e^{\beta\Delta E} - 1} \right]. \end{aligned}$$

For the "infrared" tail of the spectrum  $\Delta E \ll 1$ , the response function is

$$\mathcal{F}^{(2d)}(E) = \frac{1}{2\pi\beta} \left[ 1 - \left( \frac{7}{6} - \frac{\pi^2}{9} \right) w^2 \right] + \mathcal{O}(\Delta E) \quad (17)$$

For the "ultraviolet" tail of the spectrum  $\Delta E \gg 1$ , the excitation rate (31) has the following dominant behavior:

$$\mathcal{F}^{(a)}(E) \sim \frac{\Delta E^3}{12\pi} e^{-\beta\Delta E} \beta^2 \quad (18)$$

## B. MOTION OF THE DETECTOR IN THE THERMAL BATH

We now consider a detector moving along a spatially straight line with a constant component of its four-velocity  $\tilde{w}$

$$x^\mu(\tau) = \left( \sqrt{1 + \tilde{w}^2} \tau, 0, \tilde{w}\tau, 0 \right) \quad (19)$$

In a thermal bath (Quantum scalar field) of temperature  $T$ , corresponding to the temperature that an accelerated UnruhDeWitt detector moving along a spatially straight line with the constant magnitude of four-acceleration, trajectory (16), in Minkowski vacuum perceives, i.e.,  $T = a/(2\pi)$ .

To find any relation between  $\mathcal{F}^{(2d)}(E)$ , and the response function of the detector moving along a spatial line with constant nonrelativistic speed  $v = \tilde{w}/\sqrt{1 + \tilde{w}^2} \ll 1$  or  $w \ll 1$ . Considering a detector with the same parameters as that of the previous subsection. The Wightman thermal Green function for a detector following trajectory (16) is

$$\begin{aligned} W_\beta^+(\Delta\tau) &= -\frac{1}{4\pi^2(\Delta\tau - i\epsilon)^2} + \frac{\sqrt{1-v^2}}{8\pi\beta v \Delta\tau} \left[ \coth \left( \frac{\pi\gamma_+ \Delta\tau}{\beta} \right) \right. \\ &\quad \left. + \coth \left( \frac{\pi\gamma_- \Delta\tau}{\beta} \right) \right] + \frac{1}{4\pi^2 \Delta\tau^2} \end{aligned}$$

where  $\gamma_\pm = \sqrt{(1 \pm v)/(1 \mp v)}$ , and  $\beta = 1/T$ . The response function per unit time of this detector is

$$\mathcal{F}^{th}(E) = \frac{\sqrt{1-v^2}}{4\pi\beta v} \ln \left[ \frac{(1 - e^{-\beta\Delta E\gamma_-})}{(1 - e^{-\beta\Delta E\gamma_+})} \right] \quad (20)$$

where  $\omega = \Delta E = E - E_0$ .

For nonrelativistic velocities  $w \ll 1$ , the leading behavior of the response function is

$$\mathcal{F}^{th}(E) = \mathcal{F}^{(1d)}(E) + \mathcal{F}^v(E)\tilde{w}^2 + \mathcal{O}(\tilde{w}^4)$$

where

$$\mathcal{F}^v(E) = -\frac{e^{\beta\Delta E}\Delta E^2}{12\pi\beta[e^{\beta\Delta E}-1]^2} \left[ 3\beta^2 - \beta^3\Delta E \left( \frac{e^{\beta\Delta E}+1}{e^{\beta\Delta E}-1} \right) \right],$$

and  $\mathcal{F}^{(1d)}(E)$  is the same as in [C]

For the infrared tail of the spectrum  $\Delta E \ll 1$ , the  $\mathcal{F}^{th}$  is <sup>2</sup>

$$\mathcal{F}^{th}(E) = \frac{1}{2\pi\beta} \left[ 1 - \frac{1}{6}\tilde{w}^2 \right] + \mathcal{O}(\Delta E).$$

The expression can be mapped to the expression (17) if we relate the speed of the detector to the following transformations.

$$\tilde{w} = w\sqrt{21 - 2\pi^2}/\sqrt{3} = 0.65w.$$

The detector in the Minkowski spacetime moving along trajectory (15) should move with a higher speed  $w = 1.54\tilde{w}$  to keep up with the same excitation rate of the inertial detector in a thermal bath in the infrared limit.

For the ultraviolet tail of the spectrum  $E \gg 1$ , the excitation rate has the following dominant behavior is the same as (18):

$$\mathcal{F}^{(v)}(E) \sim \frac{\Delta E^3}{12\pi} e^{-\beta\Delta E} \beta^2$$

. Therefore, in the ultraviolet regime, the dominant modification in the response of the detector following trajectory (15) compared to the black body spectrum of Unruh radiation is the same as the dominant modification perceived by a detector moving with constant four-velocity component  $\tilde{w}$  in a thermal bath along the trajectory (19).

. Using quantum field theory arguments above, the excitation probability of an Unruh-DeWitt detector in a thermal state of a scalar field in both the rest and moving frames, arriving at the expression from (20):

$$n(\omega, T_0, v) d\omega = \frac{\omega T_0}{4\pi^2 \gamma v} \ln \left( \frac{1 - e^{-\omega\sqrt{1+v}/(T_0\sqrt{1-v})}}{1 - e^{-\omega\sqrt{1-v}/(T_0\sqrt{1+v})}} \right) d\omega$$

Where as usual, the quantities with subscript 0 are measured in the rest frame, and the ones without it are measured in the moving frame. Clearly, this expression does not have the thermal form.

$$n(\omega, T) d\omega = \frac{\omega^2}{2\pi^2 (e^{-\omega/T} - 1)} d\omega$$

for some  $T(T_0, v)$ . More concretely, Landsberg states that Since any universal relativistic transformation has to be able to deal with at least the blackbody case, we conclude that such a transformation does not exist. This point has also been raised by Sewell [16], who essentially extends the proof of Landsberg to a macroscopic thermometer. Also, one could use the argument that if classical laws can be written as  $X = Y$  in the rest frame, then a relativized version can be  $X = f(v)Y + g(v)$ , where  $f(0) = 1$  and  $g(0) = 0$ , to try to find a suitable transformation that brings to a thermal-like form.

So, until now, we have seen arguments for  $T = \gamma T_0$ ,  $T = \gamma^{-1} T_0$  and  $T = T_0$  while keeping the laws of thermodynamics unchanged in form, modifications of the laws to take into account the acceleration needed to put the system in motion, covariant forms of the laws by means of scalar or vector quantities, not defining a temperature at all... with no clear winner (although maybe a loser, the early theory by Planck). Quoting again [13],

This [the van Kampen theory] is probably the most useful formulation of relativistic thermodynamics, though by no means the correct one.

## VI. NAKAMURA'S SOLUTION: DETAILED DESCRIPTION OF THE TRANSFORMATION OF THE VOLUME

The problem with Planck's Formalism was that one was not measuring the same physical quantity in the rest and the moving reference frames. To solve this issue, Rohrlich defined the vector  $dV^\mu = v^\mu dV_0$ , but his formulation also had drawbacks. To solve these issues, Nakamura [17] defines the volume of an object as the intersection of the world-tube of the object with the 3-plane  $x^\mu w_\mu = 0$ , where  $w^\mu$  is a timelike vector that defines the direction of the three-dimensional volume in the 4-dimensional spacetime. This intersection is given by

$$V^\mu = V_0 \frac{w^\mu}{w^\nu v_\nu} \quad (21)$$

where  $v^\mu$  is the four-velocity of the volume. Energy and momentum form a four-vector now, and we can now deal with thermodynamics.

This theory is based on van Kampen's theory. He began from nonrelativistic thermodynamics, where

$$dS = \beta dU - \beta P dV \quad (22)$$

and  $U = E - p^2/2M$  for a rigid body. Differentiating this expression leads to the following generalization:

$$dU = dE - \frac{p}{M} dp \Rightarrow dU = \frac{1}{\sqrt{1-v^2}} (dE - v^i dp_i) = v^\mu dp_\mu$$

So one could finally define, in the incompressible case  $dV = 0$ ,

$$dS = \beta v_\mu dp^\mu = \beta_\mu dp^\mu \quad (23)$$

Now, given  $T = \gamma^{-1}T_0$  and 21, Nakamura proposes the following generalization to the definition of entropy for the case of homogeneous stress-energy density.

$$dS = \beta_\mu V_\nu dT^{\mu\nu} - \beta_\mu P dV^\mu \quad (24)$$

Now, using the standard definition of the stress-energy tensor.

$$T^{\mu\nu} = (P + \rho_U) v^\mu v^\nu + P\eta^{\mu\nu} \quad (25),$$

He arrives at the standard, nonrelativistic expression.

$$dS = \beta_\mu V_\nu dT^{\mu\nu} - \beta_\mu P dV^\mu = \beta V_0 d\rho_U - \beta P dV_0 \quad (26)$$

One should expect this, given that entropy is a scalar quantity.

Nakamura claims that this is actually the nonrelativistic expression, but if we try to substantiate, since in general  $V_0 d\rho_U \neq d(V_0\rho_U)$ . The reason behind this is that if we take Eq. 23 as good, then Eqs. 24 and 26 do not hold after a careful derivation. What we really have is that, in the homogeneous case,  $p^\mu = T^\mu_\nu V^\nu$ , and therefore

$$dS = \beta_\mu dp^\mu - \beta_\mu P dV^\mu = \beta_\mu T^\mu_\nu dV^\nu + \beta_\mu V^\nu dT^\mu_\nu - \beta_\mu P dV^\mu$$

By inserting the definition of the stress-energy tensor, one arrives to

$$dS = \beta_0 d(V_0\rho_U) - \beta_0 P dV_0,$$

Which has the proper differential of energy.

## VII. ARE WE LOOKING AT DIFFERENT SIDES OF THE SAME COIN?

Before concluding, let me review a recent work of B  r   and V  n [18] that claims to have solved all the issues with respect to the relativistic transformation of temperature. They present a framework from which all the different transformations can be obtained by varying the relative velocity and a new quantity, the speed of the internal energy current, which is defined. Let's try to go step by step with their derivation.

They begin with the stress-energy tensor for a general charged (in the sense not only of electric charge but any other conserved charge like color, baryon number, referred to as  $q^\mu$ ) fluid.

$$T^{\mu\nu} = (\rho_U + P) v^\mu v^\nu + P\eta^{\mu\nu} + q^\mu v^\nu + q^\nu v^\mu + \Pi^{\mu\nu}$$

Imposing  $\partial_\mu T^{\mu\nu} = 0$  amounts to

$$\begin{aligned} \partial_\mu T^{\mu\nu} = & \frac{d}{d\tau} (\rho_U v^\nu + q^\nu) + (\rho_U v^\nu + q^\nu) \partial_\sigma v^\sigma \\ & + P \left( \frac{d}{d\tau} v^\nu + v^\nu \partial_\sigma v^\sigma \right) - \frac{dv_\sigma}{d\tau} (v^\nu q^\sigma + \Pi^{\nu\sigma}) \\ & - \nabla^\nu P + \nabla_\sigma (v^\nu q^\sigma + \Pi^{\nu\sigma}) = 0 \end{aligned}$$

where  $d/d\tau = v^\mu \partial_\mu$  and  $\nabla^\mu = \partial^\mu - v^\mu v_\nu \partial^\nu$ .

Assuming  $v^\mu \partial_\mu v^\nu = 0$  (that the four-velocity does not vary), they choose a 3-dimensional surface  $S$  perpendicular to  $v$  (so that  $v \cdot s = 0 \forall s^\mu \in S$ ). This surface, given that the four-velocity does not change remains perpendicular to the four-velocity and is what they call a thermodynamic body. The idea is the same as in Nakamura (I believe). Finally, they consider homogeneous bodies ( $\nabla_\mu \rho_U = \nabla_\mu P = 0$ ), and integrate the conservation law on  $S$ .

$$\begin{aligned} \int_S dV \left[ \frac{d\rho_U}{d\tau} v^\nu + \frac{dq^\nu}{d\tau} + (\rho_U v^\nu + q^\nu) \partial_\sigma v^\sigma + p v^\nu \partial_\sigma v^\sigma \right] = \\ \int_S dV \nabla_\sigma (v^\nu q^\sigma + \Pi^{\nu\sigma}) \end{aligned}$$

Using Gauss-Ostrogradsky in the r.h.s. and Reynolds' transport theorem to the l.h.s., they arrive at

$$\frac{dE}{d\tau} \bar{v}^\mu + \frac{dW^\mu}{d\tau} + P \bar{v}^\mu \frac{dV}{d\tau} = \oint_{\partial S} (v^\mu q^\nu + \Pi^{\mu\nu}) dA_\nu = \delta Q^\mu$$

where  $\bar{v}^\mu = V^{-1} \int_S v^\mu dV$  is the average velocity inside the region  $S$ ,  $E = \rho_U V$  is the total energy and  $W^\mu = \int_S q^\mu dV$  is the total charge.

Taking a look at the middle side of the expression, we see that we have the flux of charge and energy momentum. This is the definition they give for the total heat flow  $\delta Q^\mu$ , and since we know that heat flow and entropy are related, we can consider the following relation

$$\delta Q^\mu = \frac{dE^\mu}{d\tau} + p \bar{v}^\mu \frac{dV}{d\tau} = A^\mu \frac{dS}{d\tau} + \Sigma^\mu$$

where  $E^\mu = E \bar{u}^\mu + G^\mu$  is the energy-momentum vector,  $d\bar{u}^\mu/d\tau = 0$  and  $\Sigma^\mu$  is orthogonal to  $A^\mu$ .

Multiplying this by  $A_\mu d\tau/A^2$  and taking into account that  $d\bar{v}^\mu = 0$  they arrive to

$$dS = \frac{A_\mu \bar{v}^\mu}{A^2} dE + \frac{A_\mu}{A^2} dG^\mu + P \frac{A_\mu \bar{v}^\mu}{A^2} dV$$

And they can, therefore, identify.

$$\frac{1}{T} = \frac{A_\mu \bar{v}^\mu}{A^2}, \quad \frac{g^\mu}{T} = \frac{A_\mu}{A^2}$$

Finally, noting that from the definitions  $g^\mu \bar{v}_\mu = 1$ , they arrive to the  $T$  dS equation

$$T \, dS = g_\mu dE^\mu + P \, dV$$

Now, the second important definition comes from the velocity of internal energy current.

$$w^\mu := g^\mu - \bar{v}^\mu.$$

This vector satisfies  $-w^\mu w_\mu = -g^\mu g_\mu - \bar{v}^\mu \bar{v}_\mu + 2g^\mu \bar{v}_\mu = -g^\mu g_\mu - 1 + 2 \leq 1$ . Therefore  $-1 \leq w^\mu w_\mu \leq 0$  and  $w^\mu \bar{v}_\mu = 0$ , which implies that its form is  $w^\mu = (\gamma v |\mathbf{w}|, \gamma |\mathbf{w}|)$ , with  $|\mathbf{w}| \leq 1$ .

Now, they have all the tools to analyze the problem of two bodies in relative motion and thermal contact.

In equilibrium and treating the whole system as closed,  $dE_A^\mu + dE_B^\mu = 0$ ,  $dV_A + dV_B = 0$  and  $dS(E_A^\mu, V_A) + dS(E_B^\mu, V_B) = 0$ . This amounts to require

$$\frac{g_A^\mu}{T_A} = \frac{g_B^\mu}{T_B}, \quad \frac{P_A}{T_A} = \frac{P_B}{T_B}$$

Now, they assume  $\bar{v}^\mu = \gamma(1, v)$ ,  $w^\mu = \gamma(vw, w)$  and  $q^\mu = \gamma(1 + vw, v + w)$ , and playing with the above requirements they arrive to

$$\frac{v_A + w_A}{1 + v_A w_A} = \frac{v_B + w_B}{1 + v_B w_B}$$

$$\frac{\sqrt{1 - w_A^2}}{T_A} = \frac{\sqrt{1 - w_B^2}}{T_B}.$$

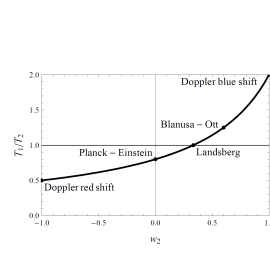
We have four velocities involved in the problem. One of them can be removed by a Lorentz transformation (essentially calculating in the rest frame of body  $A$  which will be our standing thermometer, while  $B$  will be treated as the body one wants to know the temperature of), and the first equation above gives another constraint, which can be rewritten as a function of the relative velocity  $v = (v_B - v_A) / (1 - v_B v_A)$

$$w_A = \frac{v + w_B}{1 + v w_B}$$

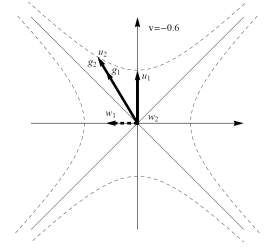
With this constraint, they recover the Doppler transformation of temperatures from the second equation of the pair.

$$T_A = T_B \frac{\sqrt{1 - v^2}}{1 + v w_B}$$

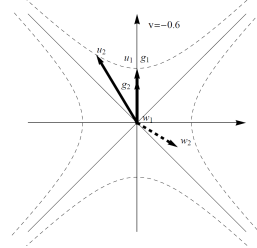
And with this formula, they managed to recover all other cases that had appeared so far.



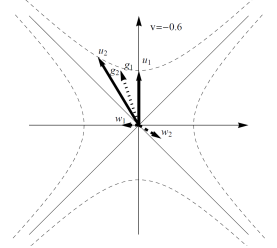
(a) Ratio of  $T_2$  (the temperatures of the observed body in its rest frame) vs.  $T_1$  (temperature shown by an ideal thermometer),  $T_1$  as a function of the speed of the heat current in the body,  $w_2$ , while approaching with the relative velocity  $v = -0.6$ .



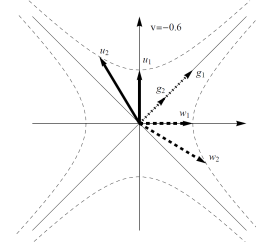
(b) Minkowski's spacetime figure for the Planck-Einstein rule of two thermodynamic bodies in equilibrium. No energy current is flowing in the observed body ( $w_2^a = 0$ ), therefore the  $u_2^a$  four-velocity (solid arrow) is parallel to the vectors ( $g_1^a, g_2^a$ ). The ratio of the temperatures is  $T_1/T_2 < 1$ .



(c) Minkowski's space-time figure for the Blanus-Ott rule. Unlike the previous case, the energy current is 0 in the thermometer ( $w_1^a = 0$ ), therefore the  $u_1^a$  four-velocity (solid arrow) is parallel to the vectors ( $g_1^a, g_2^a$ ). The ratio of the temperatures reverses, i.e.,  $T_1/T_2 > 1$ .



(d) Minkowski's spacetime figure for the Landsberg rule. There is no energy current in the composed system, therefore the four-vectors ( $g_1^a, g_2^a$ , dotted arrows) are equal. The ratio of temperatures is  $T_1/T_2 = 1$ , particularly in this case  $w_1 = -0.33$ ,  $w_2 = 0.33$ .



(e) The spacetime figure for the Doppler redshift rule of two thermodynamic bodies in equilibrium. The energy current speed (dashed arrows) in the observed body is that of light  $w_2 = 1$ , therefore  $w_1 = 1$ , and the four vectors of energy-momentum intensities ( $g_1^a, g_2^a$ , dotted arrows) are light-like.

FIG. 1. Descriptions of various spacetime figures illustrating different thermodynamic rules. Plots produced by [18]

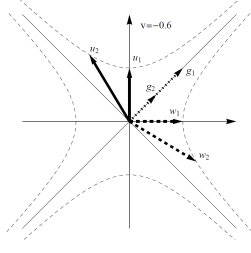


FIG. 2. The spacetime figure for the Doppler redshift rule of two thermodynamic bodies in equilibrium. The energy current speed (dashed arrows) in the observed body is that of the light  $w_2 = 1$ , therefore  $w_1 = 1$ , and the four vectors of energy-momentum intensives ( $g_1^a$ ,  $g_2^a$ , dotted arrows) are light-like. Plots produced by [18]

1. Planck-Einstein: If  $w_B = 0$ , meaning (according to them) that the energy current stands in the body, whatever that may mean, implies that  $w_A = v$ , i.e., that the energy current speed is that of the moving body, and  $T_A = \gamma^{-1}T_B$ .
2. Ott-Kibble-Møller: If  $w_A = 0$ , meaning that the current stands in the thermometer, we have that  $w_B = -v$ ,  $T_A = \gamma T_B$ .
3. Landsberg: If  $w_A + w_B = 0$ , meaning that the current is standing in the total system (something like the center-of-energy frame?), then  $w_A = (1 - \gamma^{-1})/v$  and  $T_A = T_B$ .
4. "Doppler": If  $w_B = 1$  one has a radiating body. It is also satisfied that  $w_A = 1$  and  $T_A = T_B \sqrt{\frac{1-v}{1+v}}$ .

We can also note that  $T_A = T_B$  when the relative velocity goes to zero. Finally, they give more "physically-motivated" expressions for  $g^\mu$  and  $w^\mu$

$$g^\mu = \frac{E^\mu}{E^\nu \bar{v}_\nu}$$

$$w^\mu = \frac{E^\mu - \bar{v}^\mu (E^\nu \bar{v}_\nu)}{E^\sigma \bar{v}_\sigma}$$

Interpreting this last equation as the quotient of the co-moving, average velocity-related energy current (momentum) and energy of the thermodynamic body, that is, the energy current velocity.

### VIII. CONCLUSION:

As far as we can see, the calculations become very sloppy when dealing with differentials of velocities. Sometimes we need to take them into account properly to obtain the desired result (as in Eq.  $dQ = dU + dW + v \cdot dp$ ), but sometimes it is implicitly assumed that  $dv = 0$ , which can not be clearly seen this, unless

maybe because of the restriction of no accelerations because we are in Special Relativity and not General Relativity, so velocities must be constant. If this were the case, all equations in Sections II and III hold.

On the other hand, putting equations into a covariant form does not imply the theory behind it is relativistic, as it has been shown that Newtonian mechanics can be written in a covariant form. Furthermore, in his paper, Havas [19] cites Einstein and Kretschmann when affirming that the principle of general covariance is devoid of physical content and any theory can be written in a generally covariant form.

Finally, the results by Bíró and Ván is very appealing, however, We must still work on the details and make sense of all the new construction.

### ACKNOWLEDGMENTS

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### Appendix A: Wightman Function

Considering a scalar field propagating through a Friedmann-Lemaître-Robertson-Walker (FLRW) metric

$$ds^2 = -dt^2 + a^2(t) (dr^2 + r^2 d\Omega^2), \quad (A1)$$

with spatially-flat geometry. define a new time parameter  $\eta$ , via  $d\eta = a^{-1}dt$ , conformal coordinates can be expressed like :

$$ds^2 = a^2(\eta) (-d\eta^2 + dr^2 + r^2 d\Omega^2). \quad (A2)$$

is the conformally flat nature of the FLRW metric. which we call the *conformal vacuum state* (Refs. [6, 12]. A massless scalar field  $\Phi$  with conformal coupling  $\xi = 1/6$ , characterised by the action

$$S = \frac{1}{2} \int d^4x \sqrt{-g} \left( g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi - \frac{1}{6} R \Phi^2 \right). \quad (A3)$$

Defining an auxiliary field  $\phi \equiv a\Phi$ , we may write the field equations in conformal coordinates as:

$$(\partial_\eta^2 - \partial^2) \phi = 0, \quad (A4)$$

where  $\partial^2 = \delta^{ij} \partial_i \partial_j$  is the Laplace operator and  $\phi \equiv \phi(x)$  with  $x$  indicates both temporal and spatial coordinates. Due to this choice, these field equations are invariant

under conformal transformations. Transformation of the field  $\phi$  into Fourier space yields :

$$\phi(x) = \int \frac{d^3\mathbf{k}}{(2\pi)^{3/2}} \phi_{\mathbf{k}}(\eta) e^{i\mathbf{k}\cdot\mathbf{x}}, \quad (\text{A5})$$

is used to express (A4) in terms of the time-dependent, real mode  $\phi_{\mathbf{k}} \equiv \phi_{\mathbf{k}}(\eta)$  as:

$$\phi_{\mathbf{k}}'' + \omega_{\mathbf{k}}^2 \phi_{\mathbf{k}} = 0, \quad (\text{A6})$$

$$\omega_{\mathbf{k}} = |\mathbf{k}|, \quad (\text{A7})$$

which has a general solution

$$\phi_{\mathbf{k}} = \frac{1}{\sqrt{2}} (a_{\mathbf{k}}^- v_{\mathbf{k}}^* + a_{-\mathbf{k}}^+ v_{\mathbf{k}}). \quad (\text{A8})$$

Here,  $a^\pm$  are complex constants of integration dependent only on the vector  $\mathbf{k}$  satisfying  $a_{\mathbf{k}}^+ = (a_{\mathbf{k}}^-)^*$ .  $v_{\mathbf{k}}$  are the mode functions which are normalised<sup>6</sup> such that it satisfies:

$$\phi_{\mathbf{k}} \partial_\eta \phi_{\mathbf{k}}^* - \phi_{\mathbf{k}}^* \partial_\eta \phi_{\mathbf{k}} = \frac{1}{2} \left[ v_{\mathbf{k}}' v_{\mathbf{k}}^* - v_{\mathbf{k}} v_{\mathbf{k}}'^* \right] = i. \quad (\text{A9})$$

The definition of the scalar product<sup>7</sup> results in the normalization condition:

$$(\phi_1, \phi_2) = -i \int_\Sigma d\Sigma^\mu \sqrt{-g_\Sigma} \phi_1 \overleftrightarrow{\partial}_\mu \phi_2^*, \quad (\text{A10})$$

Substitution (A9) into (A5) the mode decomposition can be written as :

$$\hat{\phi}(x) = \frac{1}{\sqrt{2}} \int \frac{d^3\mathbf{k}}{(2\pi)^{3/2}} (\hat{a}_{\mathbf{k}}^- v_{\mathbf{k}}^* e^{i\mathbf{k}\cdot\mathbf{x}} + \hat{a}_{\mathbf{k}}^+ v_{\mathbf{k}} e^{-i\mathbf{k}\cdot\mathbf{x}}), \quad (\text{A11})$$

where the constants  $a^\pm$  have been elevated to creation and annihilation operators satisfying the usual commutation relations

$$[\hat{a}_{\mathbf{k}}^-, \hat{a}_{\mathbf{k}'}^+] = \delta_{\mathbf{k}\mathbf{k}'}, \quad [\hat{a}_{\mathbf{k}}^-, \hat{a}_{\mathbf{k}'}^-] = [\hat{a}_{\mathbf{k}}^+, \hat{a}_{\mathbf{k}'}^+] = 0, \quad (\text{A12})$$

following canonical quantization, Refs. [6, 12].

and also, there exists a complete set of mode solutions  $v_{\mathbf{k}}$  satisfying

$$(v_{\mathbf{k}}, v_{\mathbf{k}'}) = \delta_{\mathbf{k}\mathbf{k}'}, \quad (v_{\mathbf{k}}^*, v_{\mathbf{k}'}^*) = -\delta_{\mathbf{k}\mathbf{k}'}, \quad (v_{\mathbf{k}}, v_{\mathbf{k}'}^*) = 0. \quad (\text{A13})$$

<sup>6</sup> In writing down the normalization condition (A9), one must take note of the following identities: As  $\phi$  is real,  $\phi^* = \phi$ ; from (A5), we have  $\phi_{\mathbf{k}}^* = \phi_{-\mathbf{k}}$ ; and as  $v_{\mathbf{k}}$  depends on  $|k|$ , we have  $v_{\mathbf{k}} = v_{-\mathbf{k}}$ . This, along with the identity  $a_{\mathbf{k}}^+ = (a_{\mathbf{k}}^-)^*$ , gives  $a_{\mathbf{k}}^+ a_{\mathbf{k}}^- = a_{-\mathbf{k}}^- a_{-\mathbf{k}}^+$ , which is required to write down (A9).

<sup>7</sup> Here,  $d\Sigma$  is the volume element of a spacelike hypersurface  $\Sigma$  (assumed to be a Cauchy surface), and we define  $d\Sigma^\mu \equiv n^\mu d\Sigma$  for a future-directed unit vector  $n^\mu$  orthogonal to the hypersurface.

The mode expansion (A11) along with the commutation relations (A12) yields quantization of the field  $\hat{\phi}$  and the normalization condition (A9), while the mode functions  $v_{\mathbf{k}}$  are specific to the theory. Because  $v_{\mathbf{k}} \equiv v_{\mathbf{k}}(\eta)$  form a basis of solutions to (A4), the explicit expressions for these can be found by solving the differential equation

$$v_{\mathbf{k}}'' + k^2 v_{\mathbf{k}} = 0 \quad \text{to find} \quad v_{\mathbf{k}} = \frac{e^{i|k|\eta}}{\sqrt{|k|}}. \quad (\text{A14})$$

Now, the two-point Wightman function

$$W(x; x') \equiv \langle 0 | \Phi(x) \Phi(x') | 0 \rangle, \quad (\text{A15})$$

We have defined the field  $\hat{\Phi}$  in terms of the auxiliary field  $\hat{\phi}$  so that  $\hat{\Phi} \equiv a^{-1} \hat{\phi}$ . Substitution of the mode decomposition (A11) gives

$$W(x; x') = \langle 0 | \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{1}{2|k|} \frac{v_{\mathbf{k}}^*(\eta) v_{\mathbf{k}}(\eta')}{a(\eta) a(\eta')} e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')} | 0 \rangle. \quad (\text{A16})$$

Referring the solution from (A14), we find

$$W_\epsilon(x; x') = \int \frac{d^3\mathbf{k}}{(2\pi)^3} \frac{e^{i|k|(\eta-\eta'-i\epsilon)+i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}}{a(\eta) a(\eta') |k|}, \quad (\text{A17})$$

the small parameter  $\epsilon > 0$  is introduced to ensure that the expression is a distribution. Finally, we evaluate the integral to obtain

$$W_\epsilon(x; x') = \frac{1}{(2\pi)^2} \frac{1}{a(\eta) a(\eta')} \frac{1}{-|\eta - \eta' - i\epsilon|^2 + |\mathbf{x} - \mathbf{x}'|^2}. \quad (\text{A18})$$

## Appendix B: Regularisation and sharp-switching

We should have an explicit regularisation for the Wightman Green function to examine the response of the detector explicitly. As the quantum field is in a state that satisfies the Hadamard condition, therefore following the Hadamard singularity structure, the response function can be written as :

$$\begin{aligned} \mathcal{F}(\omega) &= 2 \int_{-\infty}^{\infty} du \chi(u) \int_0^{\infty} ds \chi(u-s) \\ &\quad \times \left( \cos \omega s W(u, u-s) + \frac{1}{4\pi^2 s^2} \right) \\ &\quad + \frac{1}{2\pi^2} \int_0^{\infty} \frac{ds}{s^2} \int_{-\infty}^{\infty} du \chi(u) [\chi(u) - \chi(u-s)] \\ &\quad - \frac{\omega}{4\pi} \int_{-\infty}^{\infty} du \chi^2(u), \end{aligned} \quad (\text{B1})$$

Refs. [4, 13].  $W(u, u-s)$  is the limit of the Wightman Green function for the scalar wave equation with  $\epsilon \rightarrow 0^+$ .  $W(u, u-s)$  is regular everywhere except when  $x(u) = x(u-s)$ , or equivalently when  $s = 0$ , but this

Hadamard singularity is regularised by the term  $1/(4\pi^2)$ , so the response function is well-defined, and we are free to take the limit  $\epsilon \rightarrow 0^+$  prior to integration.

To obtain the sharp-switching limit of the response function, following Ref. [4] by considering a switching function of the form

$$\chi(u) = h_1 \left( \frac{u - \tau_0 + \delta}{\delta} \right) \times h_2 \left( \frac{-u + \tau + \delta}{\delta} \right), \quad (\text{B2})$$

for some initial (proper) time  $\tau_0$  with  $\tau > \tau_0$ . It can be stipulate that  $\delta > 0$ , while  $h_1$  and  $h_2$  are smooth functions satisfying

$$\begin{aligned} h_i(x) &= 0, & \text{for } x &\leq 0 \\ h_i(x) &= 1, & \text{for } x &\geq 1. \end{aligned} \quad (\text{B3})$$

Let's consider a detector to understand the choice of switching function, initially in an off position, indicating that the detector will not register any interaction with the quantum field and no particles will be detected. The detector will begin to register interaction when it is switched on smoothly via  $h_1$  beginning at  $\tau_0 - \delta$  for a duration of  $\delta$  where it stays on for a detection time of  $\Delta\tau \equiv \tau - \tau_0$ . Then, at time  $\tau$ , the function  $h_2$  begins to smoothly switch the detector off, again for a duration of  $\delta$ , so that the detector returns to the off position at  $\tau + \delta$ . The sharp switching limit For a fixed  $\Delta\tau$ , is given by  $\delta \rightarrow 0$  Ref.[23].



FIG. 3. "The on/off switching behaviour of the function  $\chi(u)$  according to the definition and constraints given by (B2) and (B3). The detector begins in an 'off' position before being switched on smoothly, beginning at time  $\tau_0 - \delta$  for a duration of  $\delta$ . The dashed pink line indicates this smooth switching behavior, which tends towards a sharp switching limit as  $\delta$  approaches zero. After the detector is fully switched on at time  $\tau_0$ , it remains in an 'on' position, until time  $\tau$ , when the detector is smoothly turned off over a duration of  $\delta$ ." [Plot produced by :[2]]

Following the Ref. [4], the response function at the sharp-switching limit to be given by

$$\begin{aligned} \mathcal{F}(\omega) &= \frac{1}{2\pi^2} \int_{\tau_0}^{\tau} du \int_0^{u-\tau_0} ds \left( \frac{\cos \omega s}{a(u)a(u-s)(\Delta x)^2} + \frac{1}{s^2} \right) \\ &+ \frac{1}{2\pi^2} \ln \left( \frac{\Delta\tau}{\delta} \right) - \frac{\omega}{4\pi} \Delta\tau + C_1 + \mathcal{O} \left( \frac{\delta}{\Delta\tau} \right), \end{aligned} \quad (\text{B4})$$

$C_1$  is a constant. Here, the perturbation is around the

small parameter  $\delta/\Delta\tau$  where  $\delta$  is the interval of smooth-switching.<sup>8</sup>

$$\begin{aligned} \dot{\mathcal{F}}_{\tau}(\omega) &= \frac{1}{2\pi^2} \int_0^{\Delta\tau} ds \left( \frac{\cos \omega s}{a(u)a(u-s)(\Delta x)^2} + \frac{1}{s^2} \right) \\ &+ \frac{1}{2\pi^2 \Delta\tau} - \frac{\omega}{4\pi} + \mathcal{O} \left( \frac{\delta}{(\Delta\tau)^2} \right), \end{aligned} \quad (\text{B5})$$

Thus, the transition rate for a detector traveling through an FLRW spacetime is given by the following:

$$\dot{\mathcal{F}}_{\tau}(\omega) = \frac{1}{2\pi^2} \int_0^{\Delta\tau} ds \left( \frac{\cos \omega s}{\sigma^2(\tau, s)} + \frac{1}{s^2} \right) + \frac{1}{2\pi^2 \Delta\tau} - \frac{\omega}{4\pi}, \quad (\text{B6})$$

Here the two-point Wightman function is written in terms of the proper time parameters  $u = \tau$  and  $s$ , with the latter encoding the history of the detector, while  $\sigma^2(\tau, s) \equiv a(\tau)a(\tau-s)(\Delta x)^2$  indicates the geodesic distance.

### Appendix C: Unrun Detector and Thermal Green Function for mass-less scalar field:

Suppose we have a pointlike two-level system (detector) moving along a worldline described by the functions  $x^\mu(\tau) = (t(\tau), \mathbf{x}(\tau))$ , where  $\tau$  is the detector's proper time, and  $\mu$  labels the coordinates in the spacetime. Assuming that this two-level system has two internal energy levels labeled by the energy  $E_0$  and  $E > E_0$  and is coupled to a quantum scalar field  $\varphi$  via a monopole interaction,  $V = mq(\tau)\varphi[x(\tau)]$ , where  $q(\tau)$  is the monopole

moment operator, and  $m$  is the interaction constant. The Hamiltonian describes the two-level detector and the quantum field :

$$\hat{H} = \hat{H}_0^{(o)} + \hat{H}_0^{(f)} + \hat{V}$$

Where  $\hat{H}_0^{(o)}$  is the Hamiltonian of the free two-level system,  $\hat{H}_0^{(f)}$  is the Hamiltonian of the free quantum scalar field, and  $\hat{V}$  is the interaction term. Let's say that without interaction,  $|A\rangle$ 's are the eigenvectors of the orthonormal basis of the Hilbert space of the states of the system,

<sup>8</sup> For large detection time  $\Delta\tau$ . and finite  $\delta$  the perturbation (B4) is equally valid and the limit  $\delta \rightarrow 0$  assumes that the detection time  $\Delta\tau$  is finite.

<sup>9</sup> a detector moving with constant velocity, i.e., with zero acceleration in a Minkowski vacuum, perceives no temperature.

<sup>10</sup> In general, trajectory of the system of the two-level detector and the field will not always remain in its ground state  $E_0^{(of)}$ , but will undergo a transition to an excited state  $E^{(of)} > E_0^{(of)}$ .

$$\hat{H}_0^{(of)}|A\rangle = E_A^{(of)}|A\rangle, \quad \hat{H}_0^{(of)} = \hat{H}_0^{(o)} + \hat{H}_0^{(f)}$$

For very small interaction constant  $m$ , in the first-order approximation of the perturbation theory, the probability amplitude of the transition from the initial state  $|A\rangle$  to the final state  $|B\rangle$  at the proper time  $\tau$  is given by following

$$\mathcal{A}_{BA} = -im \int_{-\infty}^{\tau} d\tau' V_{BA}(\tau')$$

$$V_{BA} = \langle B | \exp(i\hat{H}_0^{(of)}\tau) \hat{\varphi}(0)\hat{q}(0) \exp(-i\hat{H}_0^{(of)}\tau) | A \rangle$$

Define the states  $|n\rangle$  and  $|N\rangle$  as the eigenstates of the non-interacting free Hamiltonian of the two-level detector and the non-interacting free Hamiltonian of the free field,

$$\hat{H}_0^{(o)}|n\rangle = E_n|n\rangle, \quad \hat{H}_0^{(f)}|N\rangle = \omega_N|N\rangle,$$

Respectively. Then, the states  $|A\rangle$  of the free system of the two-level detector and field can be written as

$$|A\rangle = |n\rangle|N\rangle$$

The two-level detector is either in the ground state  $|0\rangle$  with energy  $E_0$  or in the excited state  $|1\rangle$  with energy  $E$ . The probability amplitude of the transition on the basis of  $n$  and  $N$  can be written as

$$V_{BA} = V_{MmNn} = q_{mn} e^{i(E_m - E_n)\tau} \langle M | \hat{\varphi}[x(\tau)] | N \rangle,$$

$$q_{mn} = \langle m | q(0) | n \rangle$$

Indicating the subscript  $M$  for Minkowski vacuum, the field  $\varphi$  is initially in vacuum state  $|0_M\rangle$ , and the two-level system is in ground state  $E_0$ . Considering another copy of the two-level Unruh-DeWitt detector, where these copies are different only in the value of their second energy level  $E$ . Assume that all of these detectors are prepared in the same initial state and following identical trajectories. In this ensemble of the detectors, the transition probability to all possible  $|M\rangle$  and  $|1\rangle$  's (of different values of energy  $E$ ) is given by :

$$p_{M100} = m^2 \sum_E |q_{10}|^2 \int_{-\infty}^{\tau} d\tau' \int_{-\infty}^{\tau} d\tau'' e^{i(E - E_0)\Delta\tau} \times W^+(x(\tau'), x(\tau'')) ,$$

$W^+$  indicates the positive frequency Wightman function,

$$W^+(x(\tau'), x(\tau'')) = \langle 0 | \hat{\varphi}[x(\tau')] \hat{\varphi}[x(\tau'')] | 0 \rangle ,$$

We know that massless scalar fields give

$$W^+(x(\tau'), x(\tau'')) = -\frac{1}{4\pi^2 \left[ (t' - t'' - i\epsilon)^2 - |\mathbf{x}' - \mathbf{x}''|^2 \right]}$$

In this case,  $\epsilon \ll 0$ . Here  $(t', \mathbf{x}')$  and  $(t'', \mathbf{x}'')$  are functions of the proper time. If  $W^+(x(\tau'), x(\tau''))$  can be written as  $W^+(\Delta\tau, r)$ , where  $r = |\mathbf{x}' - \mathbf{x}''|$  and  $\Delta\tau = \tau' - \tau''$ , the integrand  $p_{M100}$  depends only on  $\Delta\tau$ , can be written in the form :

$$p_{M100} = m^2 \sum_E |q_{10}|^2 \int_{-\infty}^{\tau} d\tau' \int_{-\infty}^{\infty} d(\Delta\tau) e^{i(E - E_0)\Delta\tau} \times W^+(\Delta\tau, r).$$

The transition probability per unit of proper time can be written as :

$$p_{\Delta\tau} = \frac{dp_{M100}}{d\tau} = m^2 \sum_E |q_{10}|^2 \mathcal{F}(E)$$

where

$$\mathcal{F}(E) = \int_{-\infty}^{\infty} d(\Delta\tau) e^{i(E - E_0)\Delta\tau} W^+(\Delta\tau, r),$$

It is defined as the response function per unit proper time, which is **independent of the detailed structure of the detector**. The response function of this detector per unit proper time can be calculated using the expression  $\mathcal{F}(E)$  if the detector is moving in Minkowski vacuum or using the expression  $\mathcal{F}_{\beta}(E)$  if the detector is coupled to the thermal quantum scalar field.

For a quantum scalar field that is initially in the thermal state rather than the Minkowski vacuum state, the response function  $\mathcal{F}$  can be rewritten as :

$$\mathcal{F}_{\beta}(E) = \int_{-\infty}^{\infty} d(\Delta\tau) e^{i(E - E_0)\Delta\tau} W_{\beta}^+(\Delta\tau, r)$$

Where  $W_{\beta}^+$  is the Wightman thermal Green function, which for the case of a massless scalar field is [14](page 439 to 448)

$$W_{\beta}^+(\Delta\tau, r) = W^+(\Delta\tau, r) + \frac{1}{4\pi^2 (\Delta\tau^2 - r^2)} + \frac{\coth[\pi(r + \Delta\tau)/\beta] + \coth[\pi(r - \Delta\tau)/\beta]}{8\pi\beta r}$$

where  $\beta = 1/T$  is the inverse temperature, and  $\Delta\tau = t' - t''$ .

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